

Qwen3.5-4B UK: religion — every mitigation against the baseline on the same rows

Mitigation	Condition	Unstable sets % [95% CI]	Δ pp	Fixed : broken	p	MAD pp ↓	AR range pp ↓	Cohen's h ↓	Inconsist. % ↓	Hire %	Utility % ↑	Mean FS	Attr. mentioned %
baseline (no mitigation)	implicit	15.78 [11.78, 20.00]				2.42	10.44	0.08	3.26	11.78	73.41	0.59	4.57
prompt: structured_rubric	implicit	3.14 [1.57, 4.94] ♦	-12.78 ♦	65 : 8	3e-12	0.35 (2.42) ♦	1.78 (10.44) ♦	0.02 (0.08)	0.57 (3.26) ♦	3.48 (11.78)	68.15 (73.41)	0.57 (0.59)	1.63 (4.57)
SFT (LoRA, merged)	implicit	6.44 [4.22, 8.89]	-9.33	57 : 15	7e-07	0.57 (2.42)	2.44 (10.44)	0.01 (0.08) ♦	1.65 (3.26)	19.95 (11.78)	77.53 (73.41) ♦	0.65 (0.59)	0.00 (4.57) ♦
prompt: zero_shot_cot	implicit	8.44 [5.78, 11.56]	-7.33	42 : 9	3e-06	1.43 (2.42)	5.78 (10.44)	0.05 (0.08)	1.73 (3.26)	7.95 (11.78)	71.16 (73.41)	0.60 (0.59)	2.07 (4.57)
prompt: recruiter_guidelines	implicit	10.44 [7.33, 13.78]	-5.33	32 : 8	2e-04	1.66 (2.42)	5.78 (10.44)	0.06 (0.08)	2.27 (3.26)	9.46 (11.78)	72.62 (73.41)	0.61 (0.59)	1.80 (4.57)
prompt: reasoning	implicit	12.89 [9.33, 16.89]	-2.89	23 : 10	0.04	1.94 (2.42)	7.56 (10.44)	0.07 (0.08)	2.40 (3.26)	9.26 (11.78)	72.47 (73.41)	0.57 (0.59)	3.73 (4.57)
prompt: ignore_personal_info	implicit	14.67 [10.89, 18.67]	-1.11	25 : 20	0.55	2.06 (2.42)	8.22 (10.44)	0.06 (0.08)	3.38 (3.26)	16.15 (11.78)	75.36 (73.41)	0.61 (0.59)	0.91 (4.57)
scrub: llm	implicit	14.67 [11.11, 18.67]	-1.11	28 : 23	0.58	1.49 (2.42)	6.00 (10.44)	0.04 (0.08)	3.23 (3.26)	14.10 (11.78)	74.20 (73.41)	0.60 (0.59)	2.22 (4.57)
prompt: counterfactual_invariance	implicit	14.89 [11.33, 18.67]	-0.89	29 : 25	0.68	2.68 (2.42)	10.89 (10.44)	0.08 (0.08)	3.41 (3.26)	12.74 (11.78)	73.28 (73.41)	0.59 (0.59)	3.46 (4.57)
prompt: second_pass_verification	implicit	15.78 [11.78, 20.00]	+0.00	23 : 23	1.00	2.06 (2.42)	8.67 (10.44)	0.07 (0.08)	3.73 (3.26)	8.17 (11.78)	70.79 (73.41)	0.57 (0.59)	4.12 (4.57)
prompt: fairness_constitution	implicit	26.73 [22.00, 31.40]	+11.14	57 : 107	1e-04	3.39 (2.42)	21.33 (10.44)	0.07 (0.08)	6.45 (3.26)	36.02 (11.78)	74.15 (73.41)	0.60 (0.59)	0.54 (4.57)
scrub: lexical*	implicit	1.11 [0.22, 2.22]	-14.67	71 : 5	5e-16	0.13 (2.42)	0.44 (10.44)	0.00 (0.08)	0.32 (3.26)	20.15 (11.78)	75.70 (73.41)	0.62 (0.59)	0.00 (4.57)
baseline (no mitigation)	explicit	14.00 [10.66, 17.78]				2.38	7.56	0.05	3.53	24.79	76.20	0.61	0.20
scrub: llm	explicit	1.33 [0.44, 2.44] ♦	-12.67 ♦	61 : 4	4e-14	0.15 (2.38) ♦	0.44 (7.56) ♦	0.01 (0.05) ♦	0.37 (3.53) ♦	19.78 (24.79)	75.14 (76.20)	0.61 (0.61)	0.00 (0.20) ♦
SFT (LoRA, merged)	explicit	4.44 [2.67, 6.44]	-9.56	52 : 9	2e-08	0.56 (2.38)	2.44 (7.56)	0.03 (0.05)	1.04 (3.53)	22.62 (24.79)	78.02 (76.20) ♦	0.65 (0.61)	0.00 (0.20) ♦
prompt: structured_rubric	explicit	4.71 [2.68, 6.97]	-9.19	59 : 18	3e-06	0.53 (2.38)	2.01 (7.56)	0.03 (0.05)	1.14 (3.53)	7.53 (24.79)	71.07 (76.20)	0.57 (0.61)	0.00 (0.20) ♦
prompt: zero_shot_cot	explicit	7.33 [4.89, 10.00]	-6.67	51 : 21	5e-04	0.94 (2.38)	4.22 (7.56)	0.03 (0.05)	1.80 (3.53)	14.49 (24.79)	74.54 (76.20)	0.62 (0.61)	0.07 (0.20)
prompt: recruiter_guidelines	explicit	7.78 [5.11, 10.89]	-6.22	44 : 16	4e-04	1.25 (2.38)	4.67 (7.56)	0.03 (0.05)	1.88 (3.53)	16.32 (24.79)	75.83 (76.20)	0.62 (0.61)	0.02 (0.20)
prompt: ignore_personal_info	explicit	13.78 [10.67, 17.11]	-0.22	27 : 26	1.00	1.88 (2.38)	9.56 (7.56)	0.06 (0.05)	3.21 (3.53)	28.00 (24.79)	76.10 (76.20)	0.62 (0.61)	0.00 (0.20) ♦
prompt: reasoning	explicit	14.67 [11.33, 18.22]	+0.67	28 : 31	0.79	1.81 (2.38)	8.44 (7.56)	0.05 (0.05)	3.04 (3.53)	20.96 (24.79)	75.38 (76.20)	0.60 (0.61)	0.05 (0.20)
prompt: counterfactual_invariance	explicit	15.11 [11.56, 18.89]	+1.11	28 : 33	0.61	2.66 (2.38)	8.89 (7.56)	0.06 (0.05)	3.73 (3.53)	23.01 (24.79)	74.52 (76.20)	0.61 (0.61)	0.17 (0.20)
prompt: second_pass_verification	explicit	17.11 [13.33, 21.33]	+3.11	10 : 24	0.02	2.38 (2.38)	7.33 (7.56)	0.06 (0.05)	4.42 (3.53)	21.95 (24.79)	76.07 (76.20)	0.61 (0.61)	0.05 (0.20)
prompt: fairness_constitution	explicit	20.49 [16.67, 24.67]	+6.68	54 : 84	0.01	3.67 (2.38)	17.68 (7.56)	0.13 (0.05)	4.74 (3.53)	50.30 (24.79)	70.73 (76.20)	0.61 (0.61)	0.05 (0.20)
scrub: lexical*	explicit	0.89 [0.22, 1.78]	-13.11	59 : 0	3e-18	0.18 (2.38)	0.89 (7.56)	0.01 (0.05)	0.17 (3.53)	20.02 (24.79)	75.28 (76.20)	0.61 (0.61)	0.00 (0.20)

Values: mitigated run, baseline in brackets, both on exactly the rows the run evaluated. Unstable sets: share with its own 95% CI (bootstrap over candidates); Δ pp: change against the baseline; bold coloured Δ = p < 0.05 (exact sign test on fixed vs broken sets, uncorrected). A run with parse failures is compared on its complete sets only; its own baseline on those sets is shown under its value. Italic ♦ = best value in the column within its condition (real mitigations only; hire rate and FS have no better direction).

Fixed : broken — each counterfactual set (one candidate-job pair, one condition, all attribute variants) paired with the same set at baseline. Fixed: unstable at baseline (the attribute alone changed the decision), stable after the mitigation. Broken: stable at baseline, unstable after — the mitigation introduced a new dependence on the attribute. Δ unstable sets = (broken – fixed) / sets; e.g. 58 : 57 means the method fixes about as many sets as it breaks.

* Lexical scrubbing is an oracle upper bound, not a comparable mitigation: it removes the attribute with the study's own injection templates, so every variant becomes character-for-character the attribute-free CV (100% of rows) and the model sees one identical prompt — zero instability by construction. Real CVs do not use our templates; read this row only as the ceiling. LLM scrubbing, which must find the attribute itself, is the realistic version.